MATH73235 Spring 2025 Quiz 1

First name: Alan

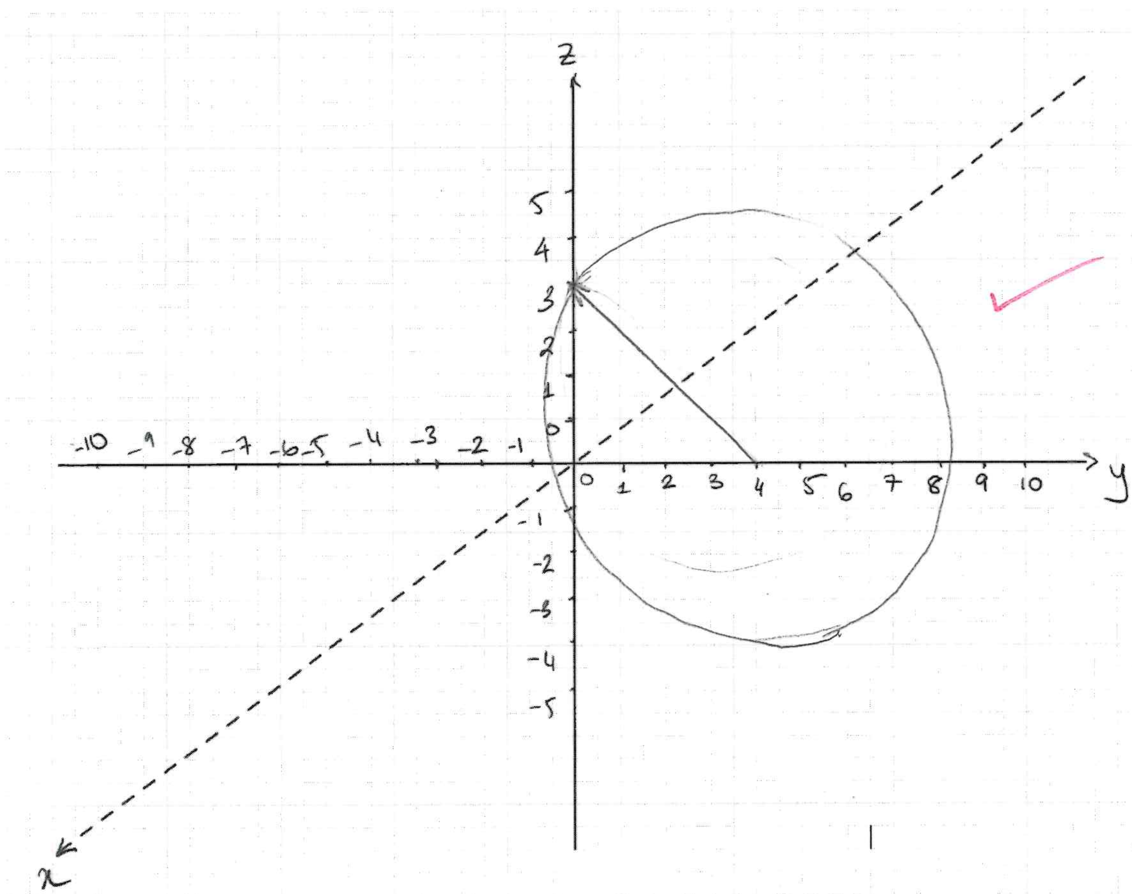
Last name: Hosseinpour moghadam

Student ID: 8825132

Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0,4,0)$ *and passing through* $(0,0,3)$. *Sketch the curve on the given axes.*

1 ***[5 marks]***

Solution. **Parametric representation:**



Sketch:

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MATH73235 Spring 2025 Quiz 1

First name: Stuart

Last name: Murray

Student ID: 8864228

5

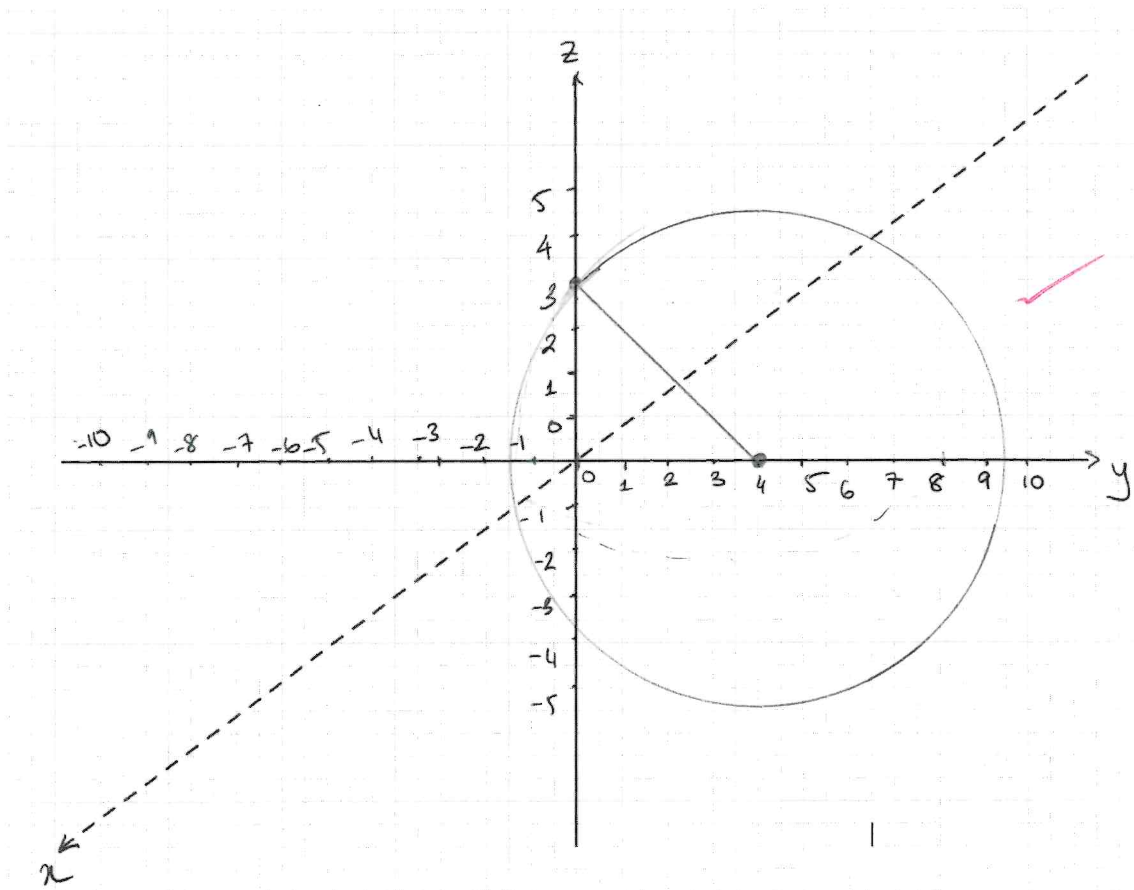
Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0, 4, 0)$ *and passing through* $(0, 0, 3)$. *Sketch the curve on the given axes.*
[5 marks]

Solution. **Parametric representation:**

$$\vec{r}(t) = \begin{bmatrix} 0 \\ 5\sin(t)+4 \\ 5\cos(t) \end{bmatrix} \in \mathbb{R}^3$$

$$q = \begin{bmatrix} 0 \\ y \\ z \end{bmatrix}$$

$$r = \sqrt{4^2+3^2} = \sqrt{25} = 5$$



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MATH73235 Spring 2025 Quiz 1

First name: AMY

Last name: WENTZELL

Student ID: 8820998

5

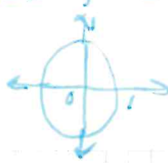
Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0, 4, 0)$ *and passing through* $(0, 0, 3)$. *Sketch the curve on the given axes.*
[5 marks]

Solution. **Parametric representation:**

$P_0 = (0, 4, 0)$

$\vec{r}(t) = y^2 + z^2$

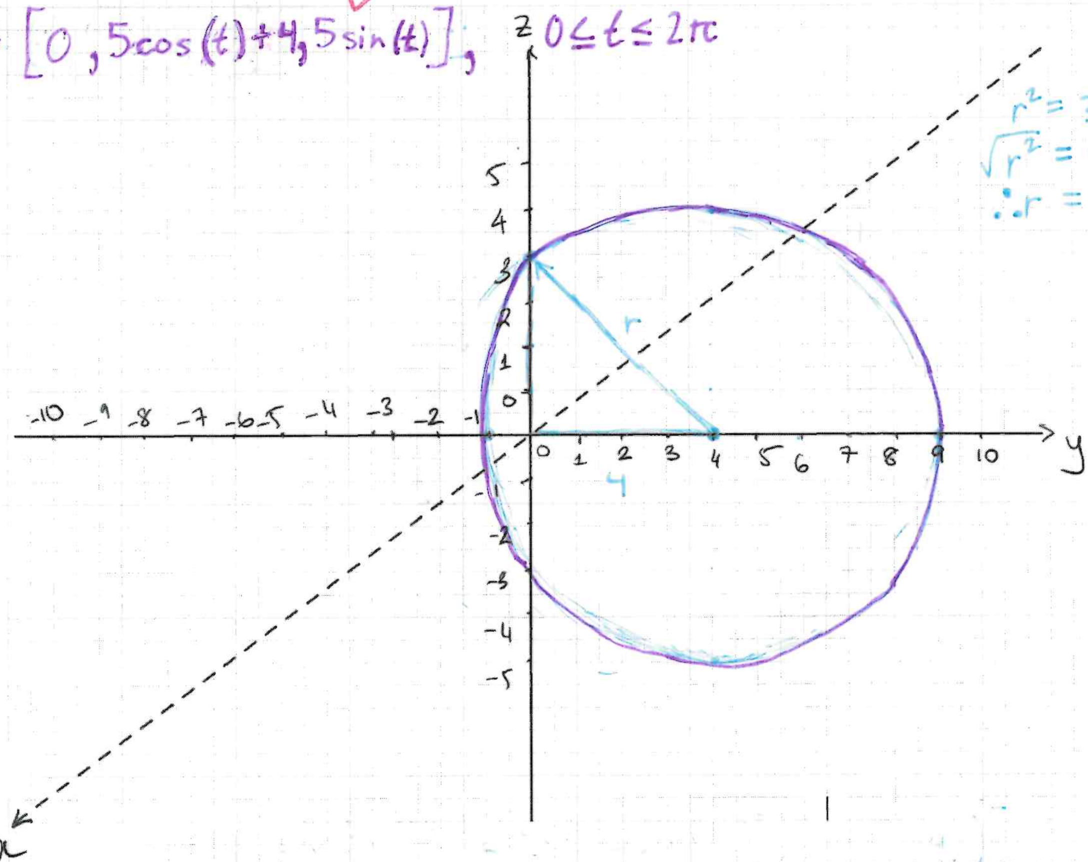
$y = \cos t, \quad z = \sin t$



$\vec{r}(t) = [0, \cos(t) + 4, \sin(t)]$

$r = 1$, BUT $P_a(0,0,3)$ exists, so:

$\therefore \vec{r}(t) = [0, 5\cos(t) + 4, 5\sin(t)], \quad 0 \le t \le 2\pi$



$r^2 = 3^2 + 4^2$

$\sqrt{r^2} = \sqrt{25}$

$\therefore r = 5$

Sketch:

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MATH73235 Spring 2025 Quiz 1

First name: David

Last name: Toma

Student ID: 8823349

4

Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0,4,0)$ *and passing through* $(0,0,3)$*. Sketch the curve on the given axes.*
[5 marks]

Solution. **Parametric representation:**

$r = \sqrt{x^2+y^2+z^2}$

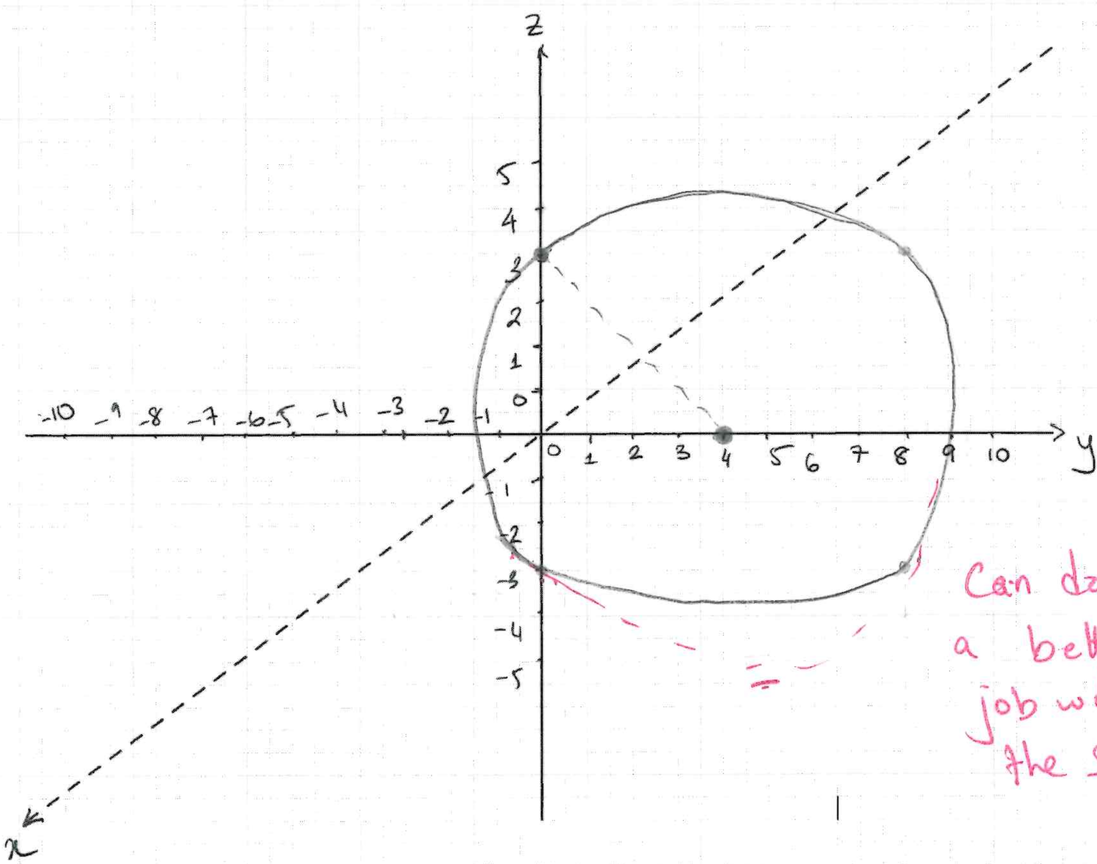
$r = \sqrt{0^2+4^2+3^2}$

$= \sqrt{16+9}$

$= \sqrt{25}$

$r = 5$

$r(t) = [0, 5\cos t + 4, 5\sin t]$



Can do a better job with the sketch

Sketch:

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MATH73235 Spring 2025 Quiz 1

First name: Coulter

Last name: Rolch

Student ID: 8751065

3

Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0,4,0)$ *and passing through* $(0,0,3)$. *Sketch the curve on the given axes.*
[5 marks]

Solution. **Parametric representation:**

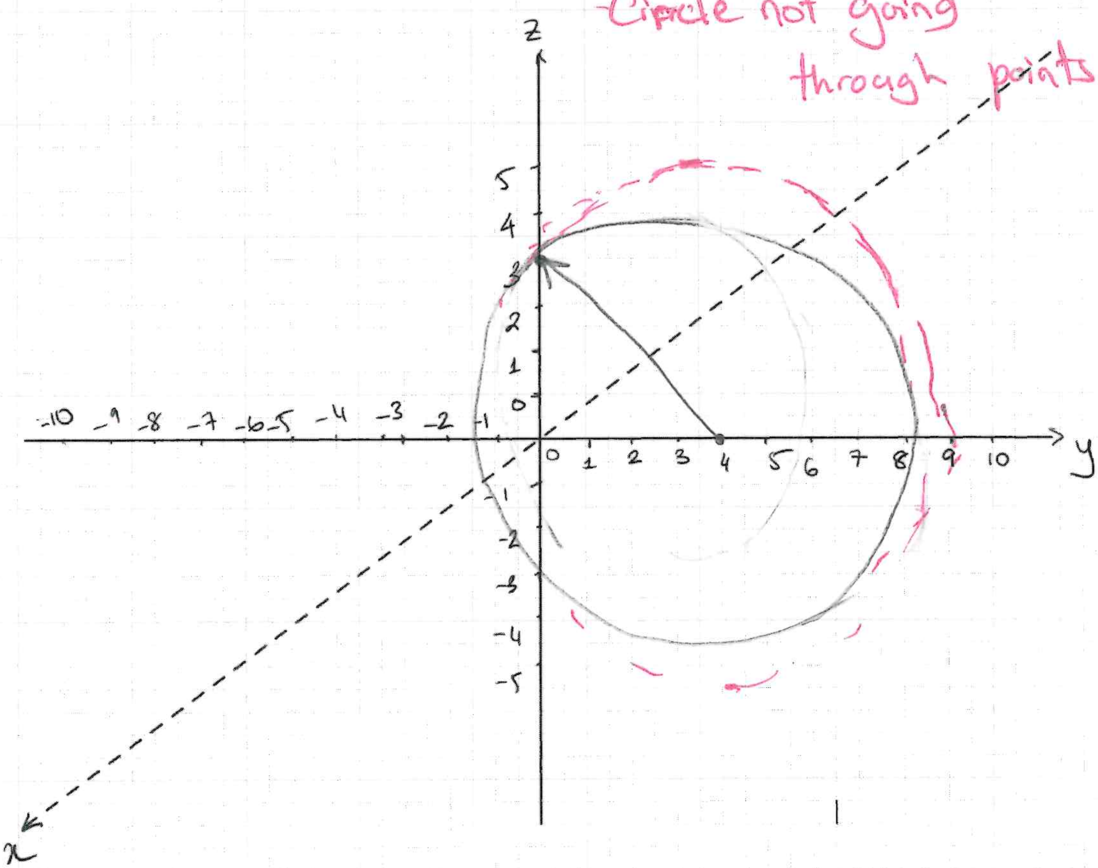
$r = \sqrt{4^2+3^2}$

$r = 5$

$\vec{r}(t) = [0, 5\sin(t)+4, \cos(t)]$

X missing radius factor

Circle not going through points



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MATH73235 Spring 2025 Quiz 1

First name: JOSHUA

Last name: DIPLOMA

Student ID: 7897960

5

Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0,4,0)$ *and passing through* $(0,0,3)$*. Sketch the curve on the given axes.*
[5 marks]

$$q = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Solution. **Parametric representation:**

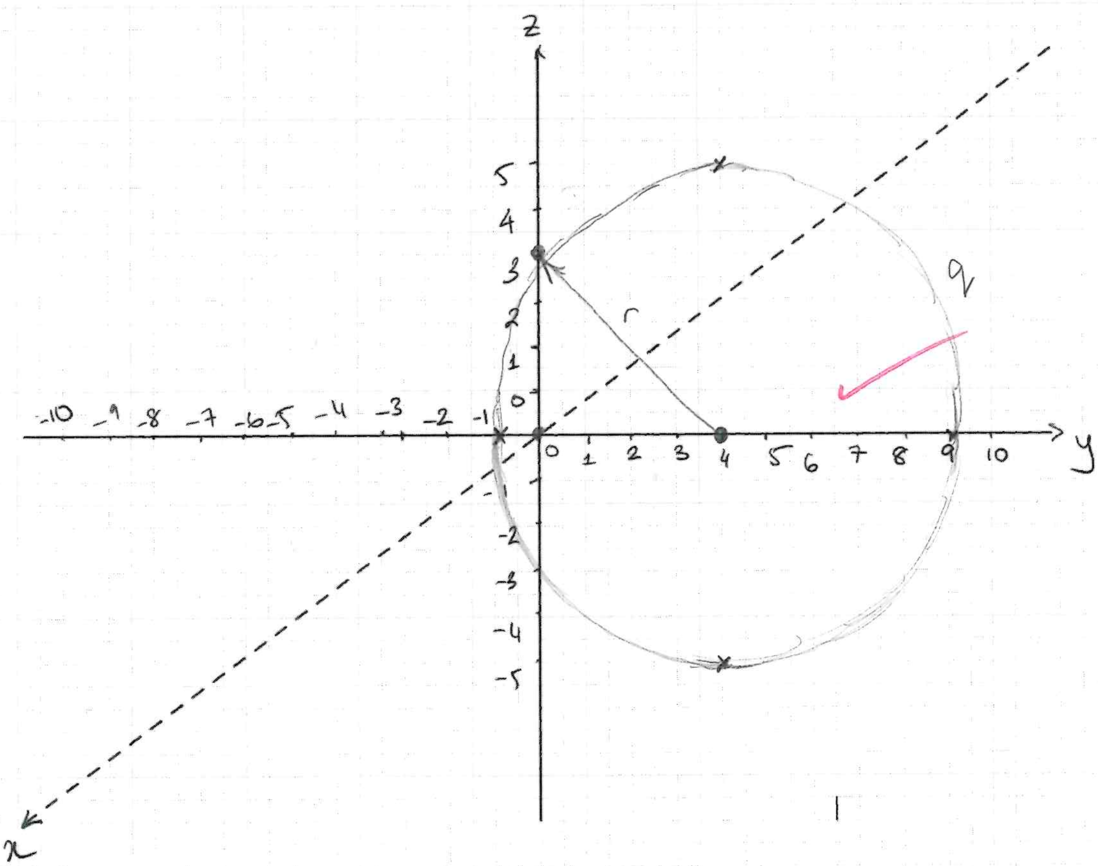
$$r = \sqrt{4^2 + 3^2}$$

$$r = \sqrt{16 + 9}$$

$$r = 5$$

$$q = \begin{bmatrix} 0 \\ 5\cos(t) + 4 \\ 5\sin t \end{bmatrix}$$

$$0 \le t \le 2\pi$$



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MATH73235 Spring 2025 Quiz 1

First name: Kyle

Last name: Dick

Student ID: 7206287

Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0,4,0)$ *and passing through* $(0,0,3)$*. Sketch the curve on the given axes.*

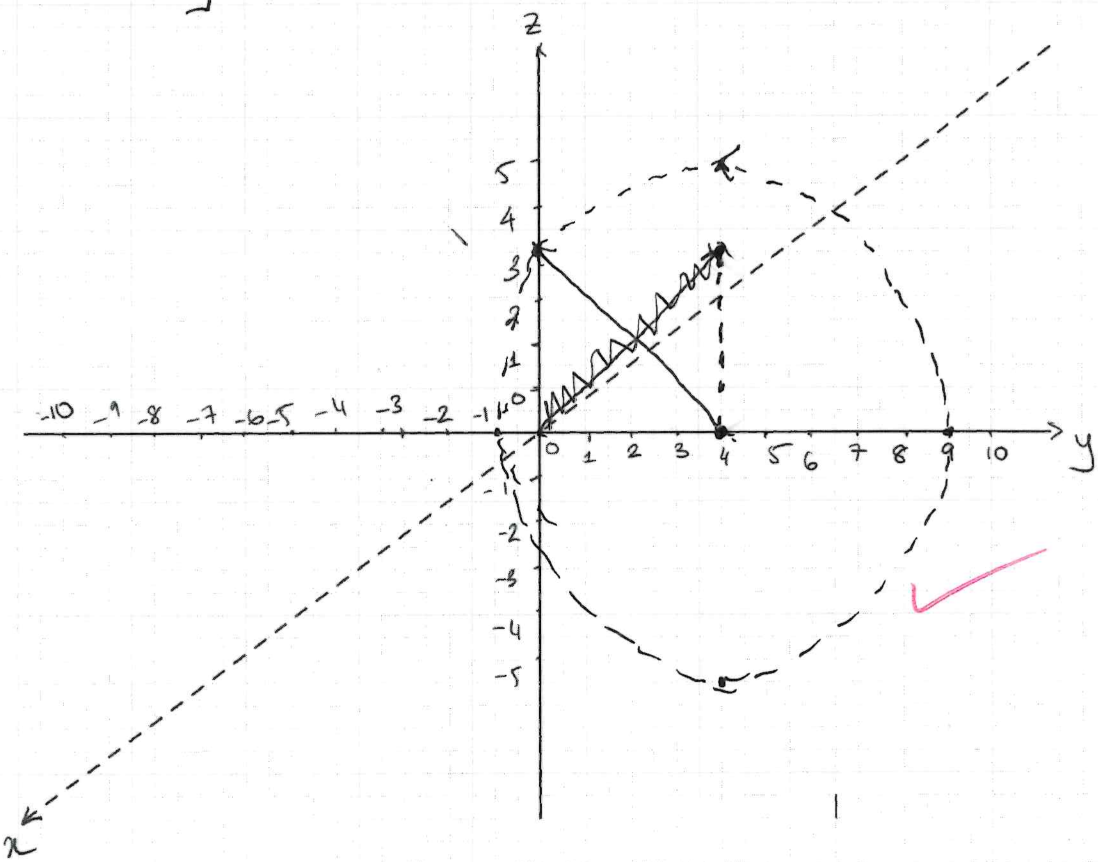
5

[5 marks]

Solution. **Parametric representation:**

$$r = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$$

$$q(t) = \begin{bmatrix} 5\cos(t) + 4 \\ 5\sin(t) \end{bmatrix} = [0,\ 5\cos(t) + 4,\ 5\sin(t)]$$



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MATH73235 Spring 2025 Quiz 1

First name: Jesse

Last name: Russell

Student ID: 7662240

Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0, 4, 0)$ *and passing through* $(0, 0, 3)$*. Sketch the curve on the given axes.*

5 ***[5 marks]***

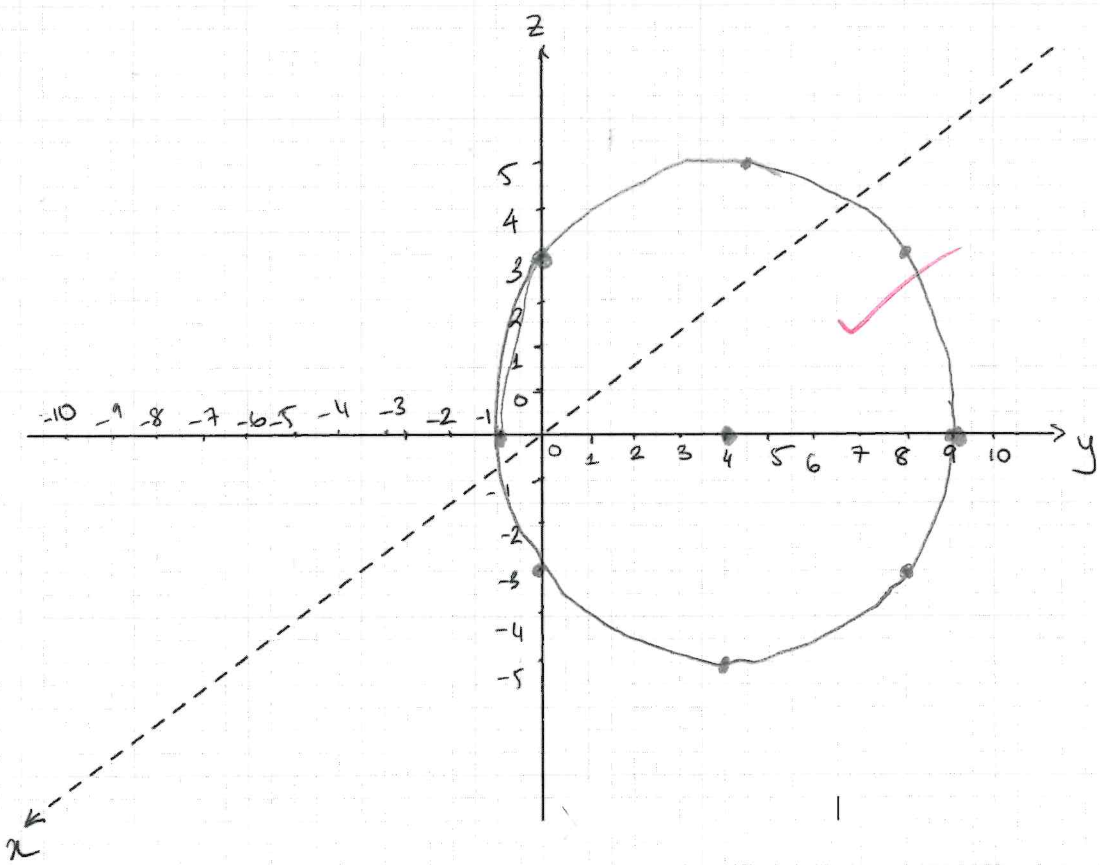
Solution. **Parametric representation:**

$r_{adius} = \sqrt{3^2 + 4^2}$

$= \sqrt{25}$

$= 5$

$[0, 5\cos(t) + 4, 5\sin(t)]$



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MATH73235 Spring 2025 Quiz 1

First name: Zach

Last name: Mcewen

Student ID: 8768637

Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0, 4, 0)$ *and passing through* $(0, 0, 3)$. *Sketch the curve on the given axes.*

4

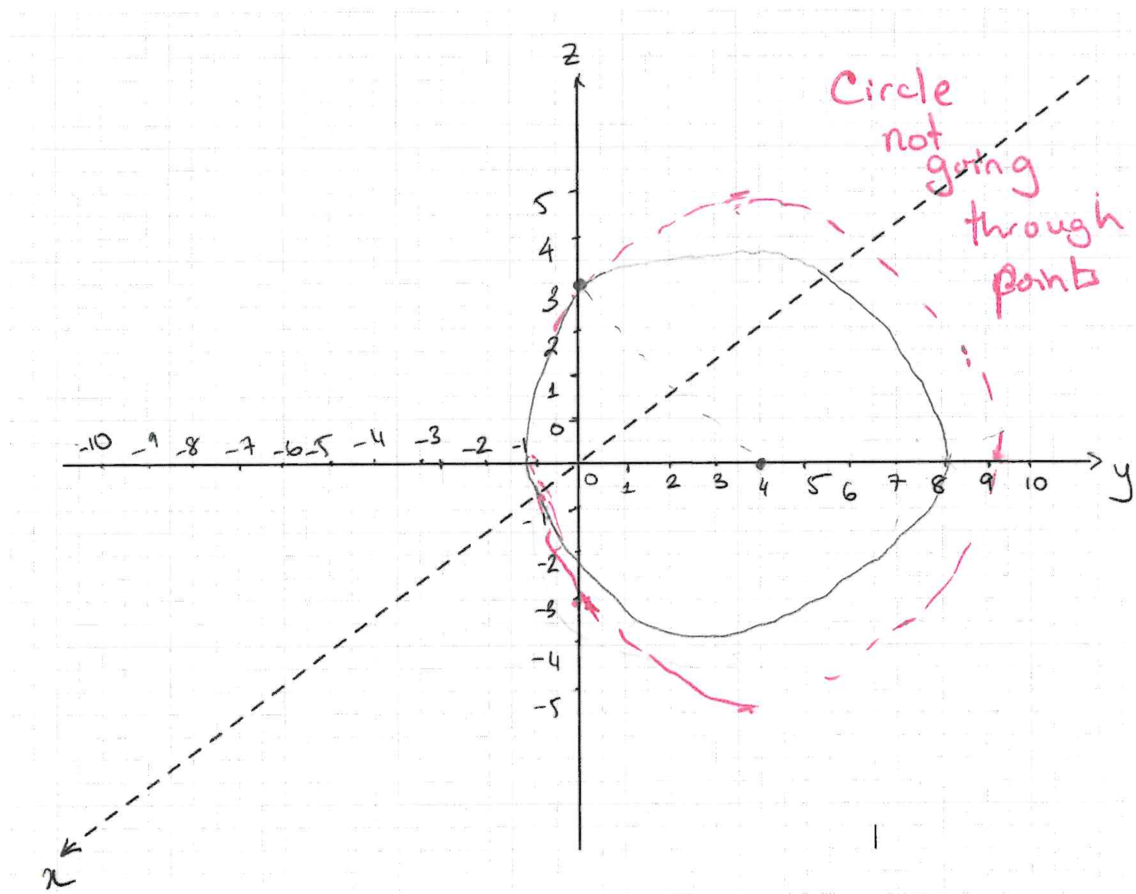
[5 marks]

Solution. **Parametric representation:**

$r = \sqrt{3^2 + 4^2}$

$r = \sqrt{25} = 5$

$\vec{r}(t) = [0, 5\sin(t) + 4, 5\cos t]$ ✓



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MATH73235 Spring 2025 Quiz 1

First name: Hayden

Last name: Foster

Student ID: 8823130

Problem 3. *Find a parametric representation of a circle in the yz-plane, with center* $(0, 4, 0)$ *and passing through* $(0, 0, 3)$. *Sketch the curve on the given axes.* Y Z

2 ***[5 marks]***

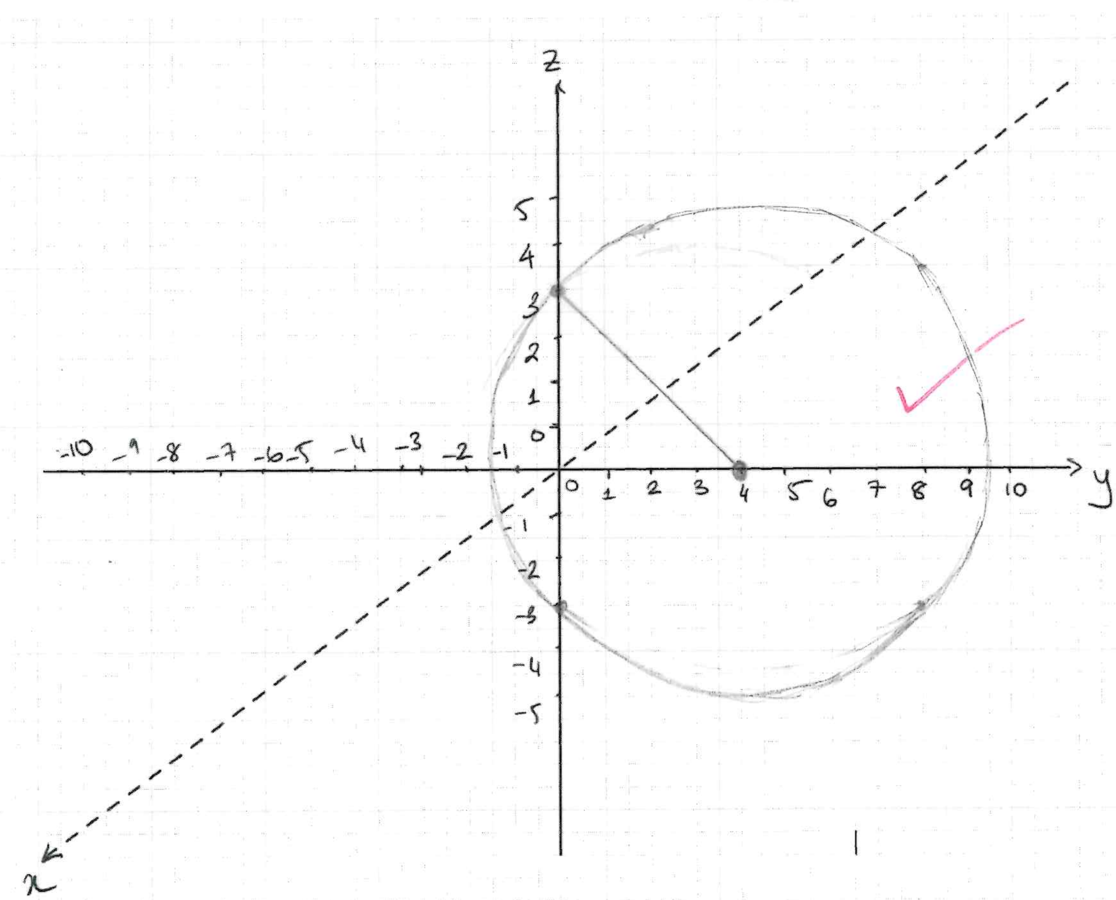
Solution. **Parametric representation:**

$r = \sqrt{3^2 + 4^2} = 5$

X

$T(x, y, z) = [0, 5(\sin(y) - 4), 5\cos(z)]$

Parametrize with a common variable such as t



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